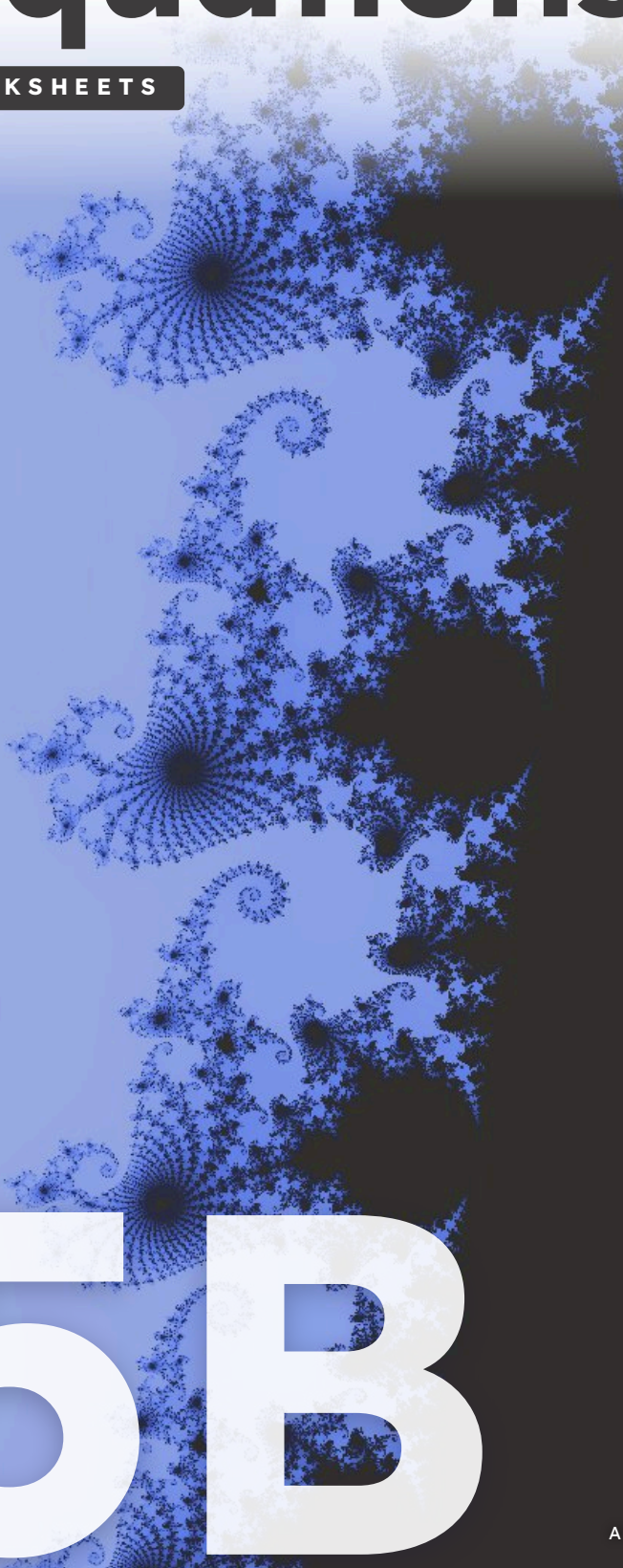


YEAR 7 MATHEMATICS

Chapter 5B

Equations

WORKSHEETS



5B

How to use this booklet

- 1** There is **one worksheet for each lesson**. It has three columns that get harder from left to right: **Easy**, then **Medium**, then **Challenging**.

- 2** Read the **speech bubble** at the top of each column: it tells you what to do. A new bubble means a new kind of question.

WE DO **Top of each column.** Your teacher works the examples on the board. **Copy every step** into the space.

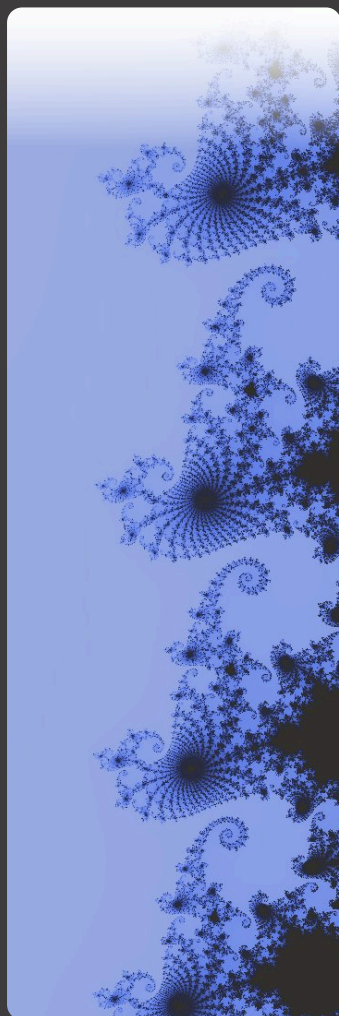
YOU DO **The rest of the column.** Your turn. Column 1: answer on the line. Columns 2 and 3: show your working on the lines.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
5.10	Guess, check and improve	3		☐
5.11	Solving one-step equations	4		☐
5.12	Solving two-step equations	5		☐
5.13	Equations and problem solving	6		☐

In this chapter you will

- check whether a value is a solution
- solve equations by guess, check and improve
- solve one-step and two-step equations using inverse operations
- write and solve equations to solve problems



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is the valley between the main body and the left bulb, where the seahorses start. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

Easy

Questions 1–30

Solve in your head: what number makes it true?

**WE DO** With your teacher: copy the working.

E1 $p + 8 = 13$

E2 $w - 11 = 9$

YOU DO Your turn.

1 $t - 9 = 3$

$t = 12$

2 $k - 6 = 4$

3 $11b = 44$

4 $b - 2 = 8$

5 $m + 2 = 15$

6 $a + 4 = 7$

7 $12w = 144$

8 $w + 12 = 25$

9 $11k = 66$

10 $t + 7 = 12$

11 $w + 6 = 9$

12 $n - 11 = 14$

13 $2m = 24$

14 $7y = 14$

15 $w - 5 = 2$

16 $6b = 78$

17 $2b = 14$

18 $9m = 54$

19 $11x = 121$

20 $p + 8 = 15$

21 $y - 8 = 2$

22 $k + 12 = 20$

23 $x + 11 = 22$

24 $k - 11 = 2$

25 $y - 3 = 4$

26 $y + 4 = 6$

27 $t - 7 = 15$

28 $x + 12 = 13$

29 $b + 5 = 20$

30 $n + 12 = 21$

Medium

Questions 31–40

Is the value a solution?
Substitute it, then compare both sides.**WE DO** With your teacher: copy the working.

E1 $3w + 1 = 22$; try $w = 7$

E2 $3p + 4 = 22$; try $p = 7$

YOU DO Your turn.

31 $5a + 6 = 16$; try $a = 2$

$5 \times 2 + 6 = 16$

$16 = 16$, so yes

32 $2y + 6 = 8$; try $y = 1$

$2 \times 1 + 6 = 8$

$8 \neq 8$, so yes

33 $4p + 7 = 15$; try $p = 1$

$4 \times 1 + 7 = 15$

$15 \neq 15$, so no

34 $6w + 2 = 14$; try $w = 2$

35 $2k + 5 = 11$; try $k = 3$

36 $3y + 1 = 4$; try $y = 2$

37 $3b + 7 = 31$; try $b = 9$

38 $4n + 9 = 17$; try $n = 2$

39 $3b + 12 = 21$; try $b = 3$

40 $2a + 1 = 5$; try $a = 2$

Challenging

Questions 41–47

Guess, check and improve.
Record each guess.**WE DO** With your teacher: copy the working.

E1 $3p + 17 = 50$

E2 $3y + 19 = 58$

YOU DO Your turn.

41 $8x + 9 = 129$

try 12: $8 \times 12 + 9 = 105$ (too small)

try 16: $8 \times 16 + 9 = 137$ (too big)

try ____: ____ $\times$ ____ + ____ = ____ ✓

42 $5y + 12 = 47$

try ____: ____ $\times$ ____ + ____ = ____ (too small)

try ____: ____ $\times$ ____ + ____ = ____ (too big)

try ____: ____ $\times$ ____ + ____ = ____ ✓

43 $6y + 12 = 84$

44 $8t + 10 = 74$

45 $8n + 3 = 99$

46 $4w + 19 = 75$

47 $5w + 18 = 73$

5.11 Solving one-step equations

YEAR 7 · CHAPTER 5B: EQUATIONS

Easy

Questions 1–30

Solve. Do the opposite operation to both sides.



WE DO With your teacher: copy the working.

E1 $a - 3 = 29$

E2 $a - 4 = 24$

YOU DO Your turn.

1 $x - 7 = 10$

$x = 17$

2 $p - 8 = 15$

3 $t + 14 = 25$

4 $t - 3 = 11$

5 $n + 18 = 33$

6 $w + 12 = 26$

7 $x + 16 = 17$

8 $b - 5 = 10$

9 $x + 8 = 12$

10 $w + 1 = 27$

11 $n - 17 = 11$

12 $t - 5 = 7$

13 $p + 17 = 31$

14 $n - 17 = 25$

15 $m + 12 = 13$

16 $w - 19 = 3$

17 $t - 8 = 26$

18 $p + 17 = 32$

19 $k - 13 = 9$

20 $y + 6 = 36$

21 $b + 10 = 25$

22 $k + 14 = 34$

23 $a + 12 = 29$

24 $n - 9 = 8$

25 $y + 3 = 21$

26 $b - 8 = 25$

27 $a + 6 = 17$

28 $n + 9 = 32$

29 $p + 14 = 35$

30 $m + 9 = 30$

Medium

Questions 31–48

Solve. Show the opposite operation on both sides.

WE DO With your teacher: copy the working.

E1 $9p = 99$

E2 $\frac{k}{7} = 5$

YOU DO Your turn.

31 $9m = 54$

32 $\frac{a}{7} = 2$

$9m \div 9 = 54 \div 9$

$\frac{a}{7} \times 7 = 2 \times 7$

$m = 6$

$a = \underline{\hspace{2cm}}$

33 $8t = 32$

34 $\frac{y}{6} = 12$

$\underline{\hspace{1cm}} t \div \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \div \underline{\hspace{1cm}}$

$t = \underline{\hspace{2cm}}$

35 $4m = 16$

36 $\frac{w}{2} = 4$

37 $7x = 84$

38 $\frac{x}{7} = 7$

39 $2n = 4$

40 $\frac{x}{2} = 11$

41 $8w = 64$

42 $\frac{n}{6} = 11$

43 $4w = 48$

44 $\frac{k}{9} = 9$

45 $8w = 16$

46 $\frac{b}{2} = 6$

47 $8n = 64$

48 $\frac{a}{6} = 12$

Challenging

Questions 49–57

Solve. The answer may be negative or a fraction.



WE DO With your teacher: copy the working.

E1 $k + 5 = -9$

E2 $7a = 6$

YOU DO Your turn.

49 $x + 2 = 1$

$x + 2 - 2 = 1 - 2$

$x = \underline{\hspace{2cm}}$

50 $10m = 28$

$\underline{\hspace{1cm}} m \div \underline{\hspace{1cm}} = \underline{\hspace{1cm}} \div \underline{\hspace{1cm}}$

$m = \frac{28}{\underline{\hspace{1cm}}}$

51 $w + 4 = -9$

52 $8w = 23$

53 $y + 6 = -1$

54 $4y = 6$

55 $t + 10 = 8$

56 $6n = 9$

57 $x + 10 = -1$

5.12 Solving two-step equations

YEAR 7 · CHAPTER 5B: EQUATIONS

Easy

Questions 1–29

Two-step equations: what is the first thing to undo? Write +, −, × or ÷ and the number.

WE DO With your teacher: copy the working.

E1 $8k - 8 = 51$

E2 $3p + 16 = 57$

YOU DO Your turn.

1 $7n - 6 = 59$

+ 6

2 $5t - 10 = 31$

3 $6w + 8 = 48$

4 $9x + 2 = 54$

5 $3p - 14 = 58$

6 $8b + 19 = 36$

7 $5k + 18 = 56$

8 $2k + 20 = 52$

9 $7t - 20 = 41$

10 $6n - 20 = 47$

11 $7p + 5 = 33$

12 $3k + 2 = 41$

13 $9y - 9 = 58$

14 $9y - 3 = 50$

15 $4p - 3 = 24$

16 $4b - 6 = 57$

17 $9w - 15 = 45$

18 $6b - 17 = 31$

19 $3t - 11 = 33$

20 $2t + 14 = 52$

21 $5a + 9 = 27$

22 $3m - 2 = 54$

23 $7t + 17 = 23$

24 $7m + 14 = 28$

25 $7t - 17 = 49$

26 $2x - 19 = 27$

27 $4w - 5 = 44$

28 $8k - 1 = 39$

29 $4a + 13 = 25$

Medium

Questions 30–51

Solve. Undo the + or − first, then the ×.

WE DO With your teacher: copy the working.

E1 $2y - 12 = -10$

E2 $9w - 5 = 31$

YOU DO Your turn.

30 $8p + 1 = 49$

31 $5b + 7 = 17$

$8p = 49 - 1 = 48$

$5b = 17 - 7 = 10$

$p = 48 \div 8 = 6$

$b = ___ \div ___ = ___$

32 $7b - 13 = 43$

33 $8k - 16 = 16$

$___ b = ___ + ___ =$

$b = ___ \div ___ = ___$

34 $3p - 15 = 21$

35 $3p - 15 = 3$

36 $8k + 19 = 67$

37 $2p - 2 = 16$

38 $8k + 19 = 115$

39 $6a + 7 = 31$

40 $2b - 17 = -1$

41 $4n - 17 = -13$

42 $8x + 12 = 28$

43 $3n - 17 = -11$

44 $8b - 6 = 58$

45 $8k + 7 = 87$

46 $6n - 5 = 31$

47 $4x + 7 = 43$

48 $3a - 19 = -16$

49 $3t - 1 = 17$

50 $8m + 12 = 92$

51 $6w + 6 = 18$

Challenging

Questions 52–58

Solve. Divide first, or expand the brackets. The answer may be negative.

WE DO With your teacher: copy the working.

E1 $6(b + 4) = 66$

E2 $9b + 5 = -31$

YOU DO Your turn.

52 $6(w + 1) = 60$

$w + 1 = 60 \div 6 = 10$

$w = 10 - 1$

$w = ___$

53 $4m + 7 = -13$

$___ m = ___ - ___ = ___$

$m = ___ \div ___$

$m = ___$

54 $6(w + 2) = 84$

55 $4b + 20 = 8$

56 $3(m + 6) = 27$

57 $5w + 7 = -28$

58 $4(k + 9) = 56$

Easy

Questions 1–18

Write an equation for each sentence. Use n for the number. (Do not solve yet.)

WE DO With your teacher: copy the working.

E1 A number plus 8 is 14. _____

E2 6 times a number is 108. _____

YOU DO Your turn.

1 A number minus 5 is 4. $n - 5 = 4$

2 A number divided by 5 is 10. _____

3 A number plus 2 is 15. _____

4 3 times a number is 18. _____

5 A number minus 9 is 20. _____

6 A number divided by 5 is 12. _____

7 A number plus 6 is 13. _____

8 5 times a number is 50. _____

9 A number minus 4 is 19. _____

10 A number divided by 3 is 6. _____

11 A number plus 3 is 20. _____

12 5 times a number is 75. _____

13 A number minus 5 is 18. _____

14 A number divided by 3 is 20. _____

15 A number plus 8 is 23. _____

16 2 times a number is 8. _____

17 A number minus 4 is 8. _____

18 A number divided by 5 is 8. _____

Medium

Questions 19–27

Write an equation, then solve it.

WE DO With your teacher: copy the working.

E1 Double a number, plus 3, is 37. _____

E2 Three times a number, minus 6, is 39. _____

YOU DO Your turn.

19 Double a number, plus 6, is 16.

$$2n + 6 = 16$$

$$n = 5$$

20 Three times a number, minus 9, is 45.

$$3n - 9 = 45$$

$$n = \underline{\hspace{2cm}}$$

21 Double a number, plus 2, is 38.

$$\underline{\hspace{1cm}}n + \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$$

$$n = \underline{\hspace{2cm}}$$

22 Three times a number, minus 5, is 19.

23 Double a number, plus 7, is 23.

24 Three times a number, minus 9, is 12.

25 Double a number, plus 4, is 22.

26 Three times a number, minus 9, is 39.

27 Double a number, plus 8, is 40.

Challenging

Questions 28–31

Write an equation for the problem, solve it, then answer in a sentence.

WE DO With your teacher: copy the working.

E1 10 equal packs hold 200 pencils. How many pencils are in each pack?

E2 A plumber charges \$55 call-out plus \$21 an hour. The bill is \$139. How many hours?

E3 Two angles of a triangle are 26° and 59° . Find the third angle, x° .

YOU DO Your turn.

28 8 equal packs hold 136 pencils. How many pencils are in each pack?

$$8p = 136$$

$$p = 136 \div 8 = 17$$

There are $\underline{\hspace{2cm}}$ pencils in each pack.

29 A plumber charges \$45 call-out plus \$26 an hour. The bill is \$149. How many hours?

$$\underline{\hspace{1cm}} + \underline{\hspace{1cm}}h = \underline{\hspace{1cm}}$$

$$\underline{\hspace{1cm}}h = \underline{\hspace{1cm}}, h = \underline{\hspace{1cm}}$$

The plumber worked $\underline{\hspace{2cm}}$ hours.

30 Two angles of a triangle are 48° and 49° . Find the third angle, x° .

31 8 equal packs hold 88 pencils. How many pencils are in each pack?